MTH 101

CALCULUS & ANALYTICAL GEOMETRY

LESSON 01.

What is calculus?

The study of continuous rates of change of quantities.

Natural numbers:-

$\{1, 2, 3, 4, \ldots\}$

Whole numbers:-

$\{0, 1, 2, 3, 4, \ldots\}$

Integers:-

$\{0, \pm 1, \pm 2, \pm 3, \ldots\}$

Rational no.:-

which can be written in form of P/q $\left[\frac{2}{3}, \frac{4}{5}\right]$

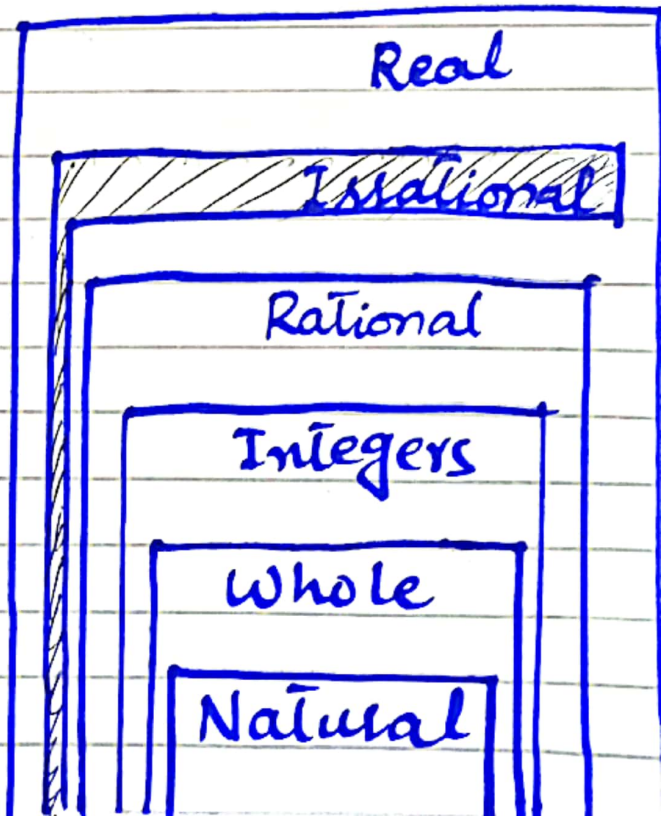
Irrational no.

which cannot be written as $\frac{p}{q}$ $(\sqrt{2}, \pi)$

Prime no.

$\{2, 3, 5, 7, 11, 13, 17, \ldots\}$

non-recurring & non-terminating numbers are called irrational numbers.



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Sets:-

=> A collection of Things

=> Always use curley brackets for representing sets.

=> The objects of a set are called elements.

=> If set A lies in set B Then A is called The subset of B. denoted as $A \subseteq B$. For example if $A = \{1, 2, 3\}$ & $B = \{1, 2, 3, 4, 5\}$ Then $A \subseteq B$.

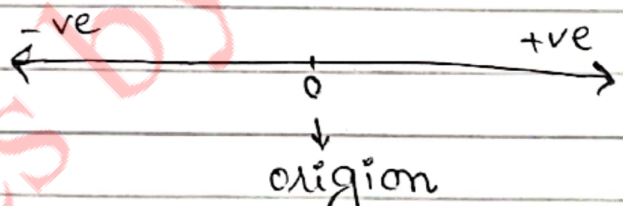
=> $\in$ symbol is used for shaving an element of a set.

For example:

$$A = \{a, b, c\}$$

Then $a \in A$ but $d \notin A$.

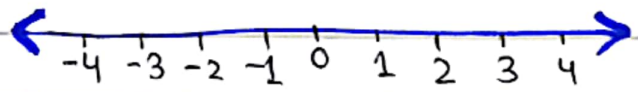
Co-ordinate line:-



The real number ~~core~~ corresponding to a point on The line is called co-ordinate of the point.

$\sqrt{2} = 1.41$

$\pi = 3.14$



Inequality:-

An expression involving $<, >, \leq, \geq, \neq$

For example $3 < 8$, $-3 > -8$

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Remark:-

$a > 0$ is a positive number.
$a \geq 0$ is non-negative

"A non-negative no. is either zero or positive."

THEOREM 1.2.1

a) If $a < b$ & $b < c$ Then $a < c$.

b) If $a < b$ & $a + c < b + c$ Then $a - c < b - c$

c) If $a < b$ & $ac < bc$ when c is +ve
& $ac > bc$ if c is -ve.

d) If $a < b$ & $c < d$ Then $a + c < b + d$

e) If a & b are both +ve or both -ve
& $a < b$ Then $\frac{1}{a} > \frac{1}{b}$

Union & Intersection of a Set

In union we combine all elements.

$B \cup A = \{1, 2, 3, 6\} \cup \{1, 2, 3, 4, 5\}$

$= \{1, 2, 3, 4, 5, 6\}$

$B \cap A = \{1, 2, 3, 6\} \cap \{1, 2, 3, 4, 5\}$

$= \{1, 2, 3\}$.

Empty set is represented by $\{\}$ or ϕ read as phi.

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Intervals:-

geometrically an interval is a line segment on the co-ordinate line.



The interval of this line segment b/w 2 & 8 will be written as

$[2, 8]$ or $\{x : 2 \leq x \leq 8\}$

2 & 8 are end points.



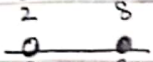
$[2, 8]$ closed interval $\{x : 2 \leq x \leq 8\}$



$(2, 8)$ open interval $\{x : 2 < x < 8\}$



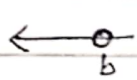
$[2, 8)$ half open $\{x : 2 \leq x < 8\}$



$(2, 8]$ half open $\{x : 2 < x \leq 8\}$



$(-\infty, b]$ closed $\{x : x \leq b\}$



$(-\infty, b)$ open $\{x : x < b\}$

Solving inequalities:-

Example:-

$$3 + 7x \leq 2x + 9$$

$$7x - 2x \leq -9 - 3$$
$$5x \leq -12$$
$$x \leq \frac{-12}{5}$$

OR

$$3 + 9 \leq 2x - 7x$$
$$12 \leq -5x$$
$$-\frac{12}{5} \geq x$$

Example:- $7 \leq 2 - 5x \leq 9$

$$7 \leq 2 - 5x$$
$$5x \leq 2 - 7$$
$$5x \leq -5$$
$$x \leq -1$$

$$2 - 5x \leq 9$$
$$2 - 9 \leq 5x$$
$$-7 \leq 5x$$
$$-7/5 \leq x$$

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$$-1 \geq x \leq \frac{-7}{5}$$

alternate:-

$$7 \leq 2-5x \leq 9$$

$$7-2 \leq 2-5x \leq 9-2$$

$$5 \leq -5x \leq 7$$

$$\frac{5}{-5} \leq -x \leq \frac{7}{5}$$

$$1 \leq -x \leq \frac{7}{5}$$

$$-1 \geq x \geq -\frac{7}{5}$$



if $-1 \geq x > -\frac{7}{5}$



LECTURE 02

Absolute value:-

denoted by $|a|$

$$|a| = \begin{cases} a & \text{if } a \geq 0 \text{ (non-negative)} \\ -a & \text{if } a < 0 \text{ (negative)} \end{cases}$$

For example

$|5| = 5$ } It is basically the no.
$|-5| = 5$ } of points moved from origion on either side.

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Example:-

$$|x-3| = 4$$

$$x - 3 = \pm 4$$

$x - 3 = 4$	$x - 3 = -4$
$x = 4 + 3$	$x = -4 + 3$
$x = 7$	$x = -1$

Example:-

$$|3x-2| = |5x+4|$$

$$(3x-2) = \pm(5x+4)$$

$3x - 2 = 5x + 4$	$3x - 2 = -(5x+4)$
$3x - 5x = 4 + 2$	$3x - 2 = -5x - 4$
$-2x = 6$	$3x + 5x = -4 + 2$
$x = -3$	$8x = -2$
	$x = -2/8$
	$x = -\frac{1}{4}$

Relation b/w Square root & absolute value:-

$$\sqrt{a^2} = |a|$$

because

$$\sqrt{a^2} = \pm a \quad , \quad |a| = \pm a$$

(e-g) $\sqrt{(7)^2} = \sqrt{(-7)^2} = \sqrt{49}$

Properties:-

$$|-a| = |a|$$

$$|ab| = |a||b|$$

$$\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$$

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Distance Formula:-

The distance b/w a & b can be found by



$$d = |b-a|$$

$$d = \begin{cases} b-a & \text{if } a<b \\ a-b & \text{if } a>b \\ 0 & \text{if } a=b \end{cases}$$

$|x-a|$ = distance b/w x & a

$|x+a|$ = distance b/w x & $-a$

$|x|$ = distance b/w x & origion

Solve:- $|x-3|<4$

$$x-3 < \pm 4$$

$x-3<4$	$x-3<-4$
$x<4+3$	$x<-4+3$
$x<7$	$x<-1$

Solve:- $|x+4| \geq 2$

$$x+4 \geq \pm 2$$

$x+4 \geq 2$	$x+4 \geq -2$
$x \geq 2-4$	$x \geq -2-4$
$x \geq -2$	$x \geq -6$

$$-2 \leq x \geq -6$$

interval notation:-

$$(-\infty, -6] \cup [-2, +\infty)$$

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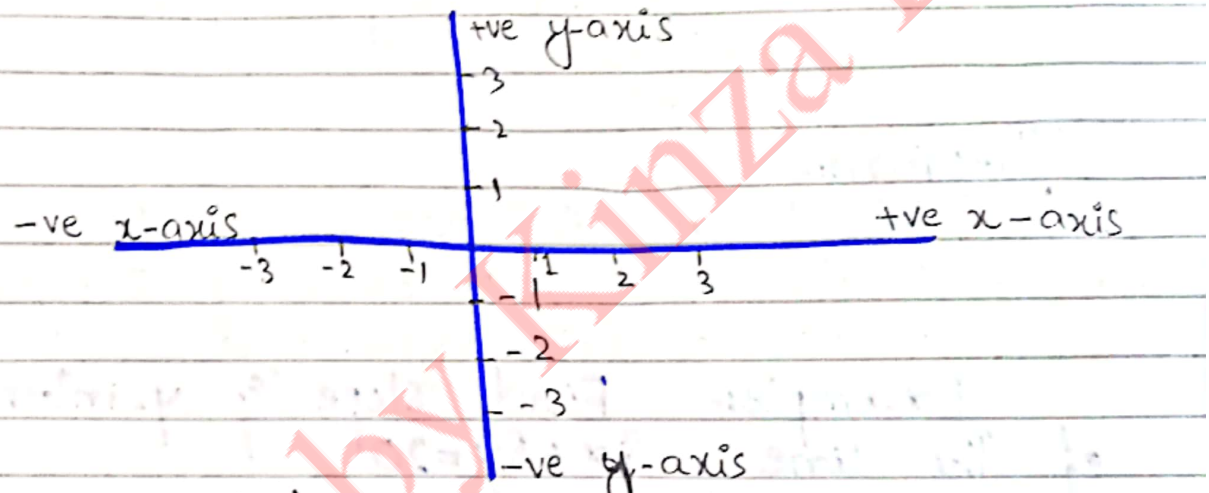
Triangle inequality:-

$$|a+b| \leq |a| + |b|$$

LECTURE 03

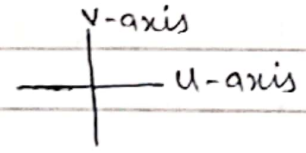
Plane:-

A Plane is intersection of two co-ordinate lines at 90 degrees. It is called a co-ordinate plane.



called xy-plane.

always write the name of horizontal axis first & then the name of y-axis (vertical a



uv-plane

Example:-

Does (3,2) satisfy $6x-4y=10$

$6x-4y=10$

put $x=3$ & $y=2$

$6(3)-4(2)=10$

$18-8=10$

$10=10$

So (3,2) is solution of $6x-4y=10$.

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Intercepts:-

In $(a,0)$ & $(0,b)$ a & b are called x-intercept & y-intercept respectively.

Example:

Find all intercepts of

a) $3x+2y=6$

For x-intercept put $y=0$

$3x+2(0)=6$

$3x=6$

$\boxed{x=2}$

For y-intercept put $x=0$

$3(0)+2y=6$

$2y=6$

$\boxed{y=3}$

b) $x=y^2-2y$

x-intercept

put $y=0$

$x=0^2-2(0)$

$\boxed{x=0}$

y-intercepts:

put $x=0$

$0=y^2-2y$

$0=y(y-2)$

$\boxed{y=0}$ or $y-2=0$

$\boxed{y=2}$

(c) $y=\frac{1}{x}$

x-intercept

$0=\frac{1}{x}$

$0\cdot x=1$

It is undefined.

y-intercept

$y=\frac{1}{0}$

$y=\infty$

It is also undefined.

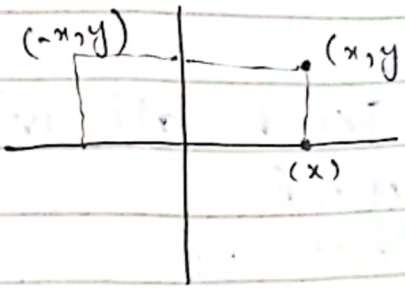
No x & y-intercepts.

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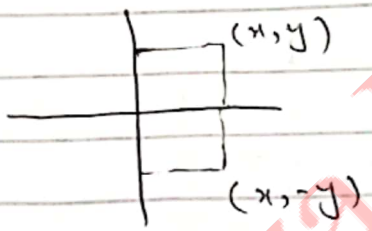
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Symmetry:-

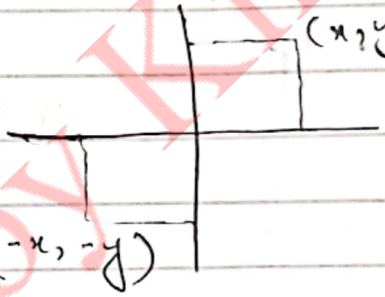
(x,y) & $(-x,y)$ are symmetric about y-axis.



(x,y) & $(x,-y)$ are symmetric about x-axis.



(x,y) & $(-x,-y)$ are symmetric about origin.



Example:-

$$y = \frac{1}{8}x^4 - x^2$$

put $x = -x$

$$y = \frac{1}{8}(-x)^4 - (-x)^2$$

$$y = \frac{1}{8}x^4 - x^2$$

As the answer is not changed so this graph will be symmetric about y-axis.

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Example:- $x = y^2$

Here if we put $y = -y$

The equation remains unchanged.

So $x = (-y)^2$

$x = y^2$

So its graph is symmetric about x-axis.

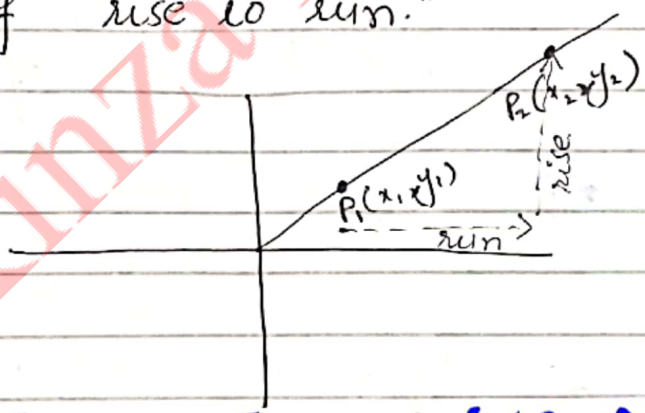
LECTURE 04

Slope:- ("rate of change of y with respect to x".)

"The ratio of rise to run."

represented by "m"

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$



Example:-

Find slope of The (i) points (6, 2) & (9, 8)

$(x_1, y_1) = (6, 2)$ & $(x_2, y_2) = (9, 8)$

$$m = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow m = \frac{8 - 2}{9 - 6} = \frac{6}{3} = 2$$

(ii) points (-2, 7) & (5, 7)

$(x_1, y_1) = (-2, 7)$ & $(x_2, y_2) = (5, 7)$

$$m = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow m = \frac{7 - 7}{5 - (-2)} = \frac{0}{5 + 2} = \frac{0}{7} = 0$$

It's formula can also be written as

$\text{rise} = m \cdot \text{run}$

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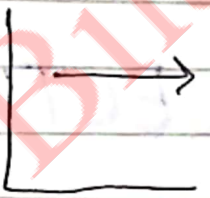
⇒ A +ve slope means that the line is inclined upward to the right. (e.g $m=2$)



⇒ A -ve slope means that the line is inclined downward to the right. (e.g $m=-3$)

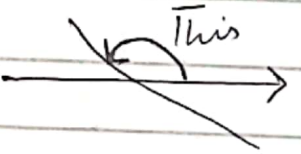


⇒ A zero slope means that the line is horizontal. (e.g $m=0$)



Angle of inclination:-

The angle of the line with +ve x-axis is called angle of inclination.



⇒ For a line parallel to x-axis this angle is always zero. ($\phi = 0$)

⇒ It's value is alway between 0° & 180°.
(i-e) $0^\circ \leq \phi \leq 180^\circ$ or $0 \leq \phi \leq \pi$

Theorem:-

For a non-vertical line, the slope m & angle of inclination ϕ are related by

$$m = \tan\phi$$

⇒ If line is parallel to y-axis (perpendicular to x-axis) Then $\phi = 90^\circ$ or $\phi = \frac{\pi}{2}$

So $\tan\phi = \tan 90^\circ =$ undefined.

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Example:-

Find angle of inclination for a line of slope $m=1$ & also for a line of slope $m=-1$

For $m=1$

$m = \tan\phi$

$1 = \tan\phi$

$\tan^{-1}(1) = \phi$

$45^\circ = \phi$

$\frac{\pi}{4} = \phi$

For $m=-1$

$m = \tan\phi$

$-1 = \tan\phi$

$\tan^{-1}(-1) = \phi$

$\frac{3\pi}{4} = \phi$

Theorem:-

Slope for parallel lines:-

The lines are parallel if & only if their slopes are same.

$$m_1 = m_2$$

Slope for perpendicular lines:-

The lines are perpendicular if & only if their slopes are negative Reciprocals of one another or the product of their slopes is always -1.

$$m_1 m_2 = -1$$

or

$$m_1 = \frac{-1}{m_2}$$

Example:-

Use slopes to show $A(1,3)$, $B(3,7)$ & $C(7,5)$ are vertices of right Triangle.

Slope of A & $B = m_1 = \frac{7-3}{3-1} = \frac{4}{2} = 2$

Slope of B & $C = m_2 = \frac{5-7}{7-3} = \frac{-2}{4} = \frac{-1}{2}$

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Here $m_1 m_2 = (2)\left(\frac{-1}{2}\right) = -1$

So the $\overline{AB}$ is perpendicular to BC.
So ABC is a right Triangle.

Vertical Line:-

In which $(a, 0)$ where $y = 0$
for example in $(-5, 0)$ $x = -5$ & $y = 0$



Horizontal Line:-

In which $(0, b)$ where $x = 0$
for example in $(0, 7)$ $x = 0$ & $y = 7$



Theorem:-

point-slope form

$$y - y_1 = m(x - x_1)$$

Example:-

Write an equation of line Through The point $(2, 3)$ with slope $\frac{-3}{2}$.

$(x_1, y_1) = (2, 3)$ & $m = \frac{-3}{2}$

using point-slope form

$$y - y_1 = m(x_2 - x_1)$$

$$y - 3 = \frac{-3}{2}(x - 2)$$

$$2y - 6 = -3(x - 2)$$

$$2y - 6 = -3x + 6$$

$$3x + 2y - 6 - 6 = 0$$

$$3x + 2y - 12 = 0$$

OR

$$2y = -3x + 12$$

$$y = \frac{-3x + 12}{2}$$

$$y = \frac{-3}{2}x + 6$$

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Example:- Write an eq. for the line through the point $(-2,-1)$ & $(3,4)$.

$$m = \frac{4-(-1)}{3-(-2)} = \frac{4+1}{3+2} = \frac{5}{5} = 1$$

we can use either of the two points as in point-slope form.

for $(3,4)$

$$y - y_1 = m(x - x_1)$$
$$y - 4 = 1(x - 3)$$
$$y - 4 = x - 3$$
$$y - x - 4 + 3 = 0$$
$$y - x - 1 = 0$$

or

$$y = x + 1$$

Theorem:-

Slope intercept form.

$$y = mx + b \quad (b \text{ is } y\text{-intercept})$$

Example:-

$$y = 2x - 5$$

here comparing with $y = mx + b$

$m = 2$ & $b = -5$

Example:

Find equation where slope is -9 & point is $(0,-4)$

here $m = -9$ & $b = -4$

$$y = mx + b$$
$$y = -9x - 4$$

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Example:-

Find slope intercept form of the eq. of line that passes through (3,4) & (-2,-1)

$$\text{Slope} = m = \frac{-1-4}{-2-3} = \frac{-5}{-5} = 1$$

Slope intercept form

$$y - y_1 = m(x - x_1)$$
$$y - (-1) = 1(x - (-2))$$
$$y + 1 = x + 2$$

General equation of a Line:-

$Ax + By + C = 0$ where A & B are non-zero.

It is also called first degree equ.

For example:- $4x + 6y - 5 = 0$

Theorem:-

Every first degree equation in x, y has a straight line graph or every straight line graph has a first degree equ.

Example:- Find slope & y-intercept of the line $8x + 5y = 20$

$$5y = -8x + 20$$
$$y = \frac{-8x}{5} + \frac{20}{5}$$
$$y = -\frac{8}{5}x + 4$$

here $m = \frac{-8}{5}$ & $b = 4$.

Slope is important bcz it gives us way to say how steep is something.

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Lecture 05

Distance Formula:-

distance between two points $A(x_1, y_1)$ & $B(x_2, y_2)$ will be

$$\boxed{d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}$$

Example:

Find distance b/w $(-2, 3)$ & $(1, 7)$.

$$d = \sqrt{[1-(-2)]^2 + (7-3)^2}$$

$$= \sqrt{(1+2)^2 + (4)^2}$$

$$= \sqrt{3^2 + 4^2}$$

$$= \sqrt{9+16}$$

$$= \sqrt{25}$$

$$\boxed{d = 5}$$

Example:- Show that the points A(4,6), B(1,-3) C(7,5) are vertices of a right triangle.

$$\overline{AB} = \sqrt{(1-4)^2 + (-3-6)^2} = \sqrt{(-3)^2 + (-9)^2} = \sqrt{9+81} = \sqrt{90}$$

$$\overline{AC} = \sqrt{(7-4)^2 + (5-6)^2} = \sqrt{3^2 + (-1)^2} = \sqrt{9+1} = \sqrt{10}$$

$$\overline{BC} = \sqrt{(7-1)^2 + (5-(-3))^2} = \sqrt{6^2 + (5+3)^2} = \sqrt{6^2 + 8^2} = \sqrt{36+64} = \sqrt{100}$$

using pythagorous theorm

$$(\overline{BC})^2 = (\overline{AB})^2 + (\overline{AC})^2$$

$$(\sqrt{100})^2 = (\sqrt{90})^2 + (\sqrt{10})^2$$

$$100 = 90 + 10$$

$$100 = 100$$

So proved that A, B, C are vertices of right triangle.

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Mid-Point Formula:-

mid point of two points $A(x_1, y_1)$, $B(x_2, y_2)$ is

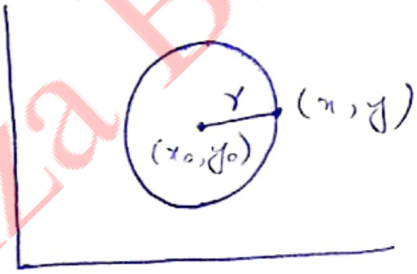
$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

Example:- Find mid-point of $(3, -4)$ & $(7, 2)$.

$$\left(\frac{3+7}{2}, \frac{-4+2}{2} \right)$$

$$= \left(\frac{10}{2}, \frac{-2}{2} \right) \Rightarrow (5, -1)$$

Circles:-



by distance formula

$$\sqrt{(x - x_0)^2 + (y - y_0)^2} = r$$

or $(x - x_0)^2 + (y - y_0)^2 = r^2$ (called standard form of equation of circle).

Example:-

Find equation of circle of radius 4 centered at $(-5, 3)$.

here $(x_0, y_0) = (-5, 3)$ & $r = 4$

using standard eqn of circle.

$$[x - (-5)]^2 + (y - 3)^2 = (4)^2$$

$$(x + 5)^2 + (y - 3)^2 = 16$$

$$x^2 + 25 + 10x + y^2 + 9 - 6y - 16 = 0$$

$$x^2 + y^2 + 10x - 6y + 18 = 0$$

Here is the equation of circle.

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Example:- Find equ of circle with center $(1,-2)$ that passes Through $(4,2)$.

using distance formula

$$r = \sqrt{(1-4)^2 + (-2-2)^2} = \sqrt{(-3)^2 + (-4)^2} = \sqrt{9+16} = \sqrt{25}$$

$$\boxed{r = 5}$$

Here $(x_0, y_0) = (1,-2)$ & $r = 5$

Put in equ of circle

$$(x-x_0)^2 + (y-y_0)^2 = r^2$$

$$(x-1)^2 + (y-(-2))^2 = r^2$$

$$(x-1)^2 + (y+2)^2 = r^2$$

$$x^2 - 2x + 1 + y^2 + 4y + 4 = (5)^2$$

$$x^2 + y^2 - 2x + 4y + 5 - 25 = 0$$

$$\boxed{x^2 + y^2 - 2x + 4y - 20 = 0}$$

Equation of circle	center	radius
$(x-2)^2 + (y-5)^2 = 9$	$(2,5)$	3
$(x+7)^2 + (y+1)^2 = 16$	$(-7,-1)$	4
$x^2 + y^2 = 25$	$(0,0)$	5
$(x-4)^2 + y^2 = 5$	$(4,0)$	$\sqrt{5}$

$x^2 + y^2 = 1$ This circle is centered at origion & has radius 1. It is called **unit circle.**

Another form of equ of circle:-

$$x^2 + y^2 + dx + ey + f = 0$$

or $Ax^2 + By^2 + Dx + Ey + F = 0$

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Example:- Find center & radius of

(i) $x^2+y^2-8x+2y+8=0$

$(x^2-8x)+(y^2+2y)+8=0$

$x^2-2(x)(4)+(4)^2-(4)^2+y^2+2(y)(1)+1^2-1^2+8=$

$(x-4)^2-16+(y+1)^2-1+8=0$

$(x-4)^2+(y+1)^2-9=0$

$(x-4)^2+(y+1)^2=9$

here $\boxed{(x_0,y_0)=(4,-1)}$ & $\boxed{r=3}$

(ii) $2x^2+2y^2+24x-81=0$

divide b/s by 2

$\frac{2x^2}{2}+\frac{2y^2}{2}+\frac{24x}{2}-\frac{81}{2}=\frac{0}{2}$

$x^2+y^2+12x-\frac{81}{2}=0$

$x^2+12x+y^2-\frac{81}{2}=0$

$x^2+2(x)(6)+(6)^2-(6)^2+y^2-\frac{81}{2}=0$

$(x+6)^2-36+y^2-\frac{81}{2}=0$

$(x+6)^2+y^2=\frac{81}{2}+36$

$(x+6)^2+y^2=\frac{153}{2}$

Here $\boxed{(x_0,y_0)=(-6,0)}$, $\boxed{r=\sqrt{\frac{153}{2}}}$

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Degenerate cases of circle:-

In $(x-x_0)^2+(y-y_0)^2=k$

($k>0$) The graph is a circle & centered at (x_0, y_0) & Radius k.

($k=0$) The only solution of the equation is $x=x_0$, $y=y_0$. So graph is a single point (x_0, y_0)

($k<0$) The equation has ~~only~~ no real graph & no real solutions.

Example:- Describe graph of

(a) $(x-1)^2+(y+4)^2=-9$

As $k<0$, so this equation has no real solution & no graph.

(b) $(x-1)^2+(y+4)^2=0$

The only values of x & y that will make the left side 0 are $(1,-4)$. So graph is a single point at $(1,-4)$.

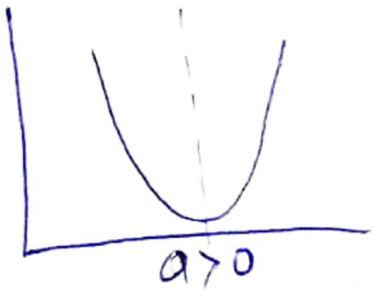
Quadratic equation:-

The equation of the form $y=ax^2+bx+c$ is called a quadratic equation.

Graph of quadratic equ.

The graph of quadratic equ is a parabola.

If a is +ve then graph is



If a is -ve then graph is



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parabola is symmetric about a line parallel to y-axis.

x-coordinate of vertex of parabola can be found as $\boxed{x = \frac{-b}{2a}}$ & vertex has highest point if $a<0$ & a lowest point if $a>0$.

Example: Find vertex of $y = -x^2+4x-5$.

$a=-1,\ b=4,\ c=-5$

$$x = \frac{-b}{2a}$$

$$x = \frac{-4}{2(-1)} \Rightarrow \frac{-4}{-2} \Rightarrow \boxed{x=2}$$

Example: (on page 55). see statement.

$$S = 24.5t - 4.9t^2$$

(a) $\text{vertex}(t) = \frac{-b}{2a} = \frac{-24.5}{2(-4.9)} = 2.5 \text{ sec}$

(b) height of the graph is max or min, depending on depending on whether the graph opens up or down.

$$S = 24.5t - 4.9t^2$$

$$S = 24.5(2.5) - 4.9(2.5)^2$$

$$S = 30.626 \text{ m}$$

So the ball rises 30.626m above the ground.

In $x = ay^2+by+c$ the vertex is $y=-b$ & graphs will be symmetric about a line parallel to x-axis.



$a>0$



$a<0$

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Lecture 06

Function: This term was used first by French Mathematician Leibiniz. He used it to denote dependence of one quantity on other.

Example:- In $A = \pi r^2$, Area of circle depends upon radius r. So Area is a function of r.

Defination: If a quantity y depends upon other quantity x in such a way that each value of x determines exactly one value of y. We say y is a function of x.

$y = 4x + 1$ is a function.

$y = \pm\sqrt{x}$ is not a function bcz y gives two values if one value of x is put.

Notations:- Euler introduced the notation $y = f(x)$ read as "y equals f of x" & y is a function of x. Here x is independent variable & y is a dependent variable.

Example:- $y = f(x) = x^2$ find $f(3)$ & $f(-2)$

$f(3) = (3)^2 \Rightarrow 9$ & $f(-2) = (-2)^2 = 4$

Remark:-

Any letter can be used instead of f.

$y = f(x)$, $y = g(x)$, $y = h(x)$ all are same

In $s = f(t)$ t is independent & s is dependent

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Example:- If $\varphi(x) = \frac{1}{x^3-1}$

find $\varphi(5^{\frac{1}{6}})$ & $\varphi(1)$

$$\varphi(5^{\frac{1}{6}}) = \frac{1}{(5^{\frac{1}{6}})^3 - 1} = \frac{1}{5^{\frac{1}{2}} - 1} = \frac{1}{\sqrt{5}-1}$$

$$\varphi(1) = \frac{1}{(1)^3 - 1} = \frac{1}{1-1} = \frac{1}{0} = \infty \text{ (undefined).}$$

Example:- $f(x) = 2x^2 - 1$, find $f(d)$ & $f(t-1)$

$$f(d) = 2d^2 - 1$$

$$f(t-1) = 2(t-1)^2 - 1 = 2(t^2+1-2t) - 1$$

$$= 2t^2 + 2 - 4t - 1 = 2t^2 - 4t + 1$$

Remark:- If two functions look alike in all aspects other than variables, then they are same. (e.g) $g(x) = x^2 - 4x$ & $g(c) = c^2 - 4c$

Domain of a function:-

The independent ~~value~~ variable is not ~~and~~ allowed to take any value.

It may be restricted to take some values. This is called domain. It is the set consisting of all elements/allowable values for independent variable.

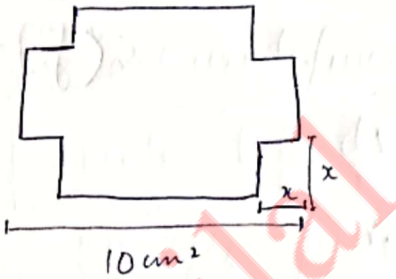
In other words "The value to be put" is called domain.

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Example:- Suppose that a square with a side of length x cm is cut from four corners of a piece of cardboard that is $10\,cm^2$ & let y be area of the cardboard that remain. Find its domain.

Solution:-



Area of cardboard = side × side

$= 10 \times 10$

$= 100\,cm^2$

area of pieces that are cut $= 4(x \times x)$

$= 4x^2$

Area of cardboard that remain $= 100 - 4x^2$

Here we cannot put values greater than 5 bcz it will give -ve answer that is not possible in the case of area. So its domain'll be

$0 \leq x \leq 5$ or $[0, 5]$.

Types of Domain:- There are two types

1. Natural Domain 2. Restricted Domain

Natural domain:- It comes out as a result of the formula of function. for example in

$h(x) = \frac{1}{(x-1)(x-3)}$ The denominator shouldn't be zero so we can't put 1 & 3 as value of x. So 1 & 3 is not part of domain. The domain is

$(-\infty, 1) \cup (1,3) \cup (3, +\infty)$

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Restricted domain:-

It is the domain which can be altered by doing some manipulations.

for example:- $h(x) = \frac{x^2-4}{x-2}$ (here we cannot put $x=2$ bcz it makes the denominator zero).

But we can do some algebric manipulation.

$$h(x) = \frac{x^2-4}{x-2} = \frac{(x)^2-(2)^2}{x-2} = \frac{(x-2)(x+2)}{x-2}$$

$$h(x) = x+2$$

Now here we can put any value of x, even 2 also. So we altered the domain.

Range of a function:-

The set of y values that come out as a result of x values are called Range.

In other words Range is set of all possible values for $f(x)$ as x varies over domain.

Techniques for finding Range:-

In $y=x^2$, the value of y alway comes out positive. So its Range is all positive real numbers.

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$\Rightarrow$ In $2+\sqrt{x-1}$ The domain will be $[1, +\infty)$

So The range will be $[2, +\infty)$

$\Rightarrow$ In $y = \frac{x+1}{x-1}$ The domain is all real numbers except 1. bcz 1 makes The denominator zero.

Solve it for x.

$$y = \frac{x+1}{x-1}$$

$$y(x-1) = x+1$$

$$xy - y = x+1$$

$$xy - x = 1+y$$

$$x(y-1) = y+1$$

$$x = \frac{y+1}{y-1}$$

So here we can see That y can also be not equal to one. So its range will be

$$(-\infty, 1) \cup (1, +\infty).$$

Functions devided piecewise:-

Sometimes The function need to be defined by formulas That have been "pieced together."

As $$f(x) = \begin{cases} 1.75 & 0 < x \leq 1 \\ 1.75 + 0.50(x-1) & 1 < x \text{ or } x > 1 \end{cases}$$

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Reversing roles of x & y:

Usually x is independent & y is dependent. But the role can be reversed for our convenience.

For example $x = 4y^5 - 2y^3 + 7y - 5$

For example

In $3x + 2y = 6$, either x or y can be seperated on our own choice.

as $3x = 6 - 2y$

$x = \frac{6}{3} - \frac{2y}{3}$

$x = -\frac{2}{3}y + 2$

$2y = 6 - 3x$

$y = \frac{6}{2} - \frac{3x}{2}$

$y = -\frac{3}{2}x + 3$

Lecture 07

Operations on functions

Example:- $f(x) = x^2$ & $g(x) = x$, find $f(x)+g(x)$

Simply add the functions as

$f(x) + g(x) = x^2 + x$

Definations:

$(f+g)x = f(x) + g(x)$

$(f-g)x = f(x) - g(x)$

$(f \cdot g)x = f(x) \cdot g(x)$

$\left(\frac{f}{g}\right)x = \frac{f(x)}{g(x)}$, $g(x) \neq 0$

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⇒ For $f+g$, $f-g$ & $f.g$ domains are intersection of domains of f & g.

⇒ For $\frac{f}{g}$ the domain is intersection of f & g except for the points where $g(x)=0$.

Example:- $f(x)=1+\sqrt{x-2}$ & $g(x)=x-1$

find domain of $(f+g)x$:-

$$(f+g)x = f(x)+g(x) = 1+\sqrt{x-2}+x-1 = x+\sqrt{x-2}$$

Domain of $f(x) = [2, +\infty)$

Domain of $g(x) = (-\infty, +\infty)$

Domain of $(f+g)x = [-2, +\infty) \cap (-\infty, +\infty)$

$= [-2, +\infty)$

Example:- $f(x)=3\sqrt{x}$ & $g(x)=\sqrt{x}$

find $f.g(x)$

$$f.g(x) = f(x).g(x)$$

$$= (3\sqrt{x})(\sqrt{x}) = 3(\sqrt{x})^2 = 3x$$

The natural domain of $f.g(x)$ is $(-\infty, +\infty)$ but restricted domain will be intersection of domains of f & g.

Dom of $f(x) = [0, +\infty)$

Dom of $g(x) = [0, +\infty)$

Dom of $f.g(x) = [0, +\infty) \cap [0, +\infty)$

$= [0, +\infty)$

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Notation:-

$f^2(x) = f(x) \cdot f(x)$

$f^3(x) = f(x) \cdot f(x) \cdot f(x)$

$f^n(x) = f(x) \cdot f(x) \cdots\cdots f(x)$.

(e.g) $(\sin x)^2 = \sin x \cdot \sin x = \sin^2 x$

Composition of functions:-

Composition has no analog with arithmatic operations.

When two functions are composed, one is assigned as a value to the independent variable of other. $f(x) = x^3$, $g(x) = x+4$

$$fog(x) = f(g(x))$$

First $f(x) = x^3$

put $x = g(x)$, $f(g(x)) = (g(x))^3$

$= (x+4)^3$

Domain of $fog(x)$ consists of all x in the domain of g for which $g(x)$ is in domain of f.

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Example:

$f(x) = x^2 + 3$, $g(x) = \sqrt{x}$

$fog(x) = f(g(x))$

First $f(x) = x^2 + 3$

$f(g(x)) = [g(x)]^2 + 3$

$= (\sqrt{x})^2 + 3$

$= x + 3$

Dom of $g = [0, +\infty)$

Dom of $f = (-\infty, +\infty)$

Dom of $fog(x) = [0, +\infty) \cap (-\infty, +\infty)$

$= [0, +\infty)$.

Remark:

$fog(x) \neq gof(x)$

Decomposing The function:-

Sometimes we decompose a function (i-e) break the function into simpler ones.

(e-g) $h(x) = (x+1)^2$

is a composition of $f(x) = x + 1$ & $g(x) = x^2$. So $h(x) = g(f(x)) = gof(x)$

Example:-

$(x^2+1)^{10} = [(x^2+1)^2]^5 = f(g(x))$, $g(x) = (x^2+1)^2$ & $f(x) = x^5$

or $(x^2+1)^{10} = [(x^2+1)^3]^{\frac{10}{3}} = f(g(x))$, $g(x) = (x^2+1)^3$ & $f(x) = x^{\frac{10}{3}}$

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Example:- $T(x) = \sqrt{\left(\frac{x}{3}\right)^3} = f(g(h(x)))$

$f(x) = \sqrt{x}$, $g(x) = x^3$, $h(x) = \frac{x}{3}$

Some more examples

Function	g(x) inside	f(x) outside	composition
$(x^2+1)^{10}$	x^2+1	x^{10}	$(x^2+1)^{10} = f(g(x))$
$\sin^3 x$	$\sin x$	x^3	$\sin^3 x = f(g(x))$
$\frac{1}{x+1}$	$x+1$	$\frac{1}{x}$	$\frac{1}{x+1} = f(g(x))$
$\tan x^5$	x^5	$\tan x$	$\tan x^5 = f(g(x))$

Constant function:-

These assign the same number to every x in the domain. (e.g) $f(x) = 2$, $f(-2) = 2$, $f(t) = 2$

Monomial in x:-

Anything like cx^n (c is constant & n is any non-negative integer) e.g. $2x^5$, $\sqrt{3}x^{55}$

$4x^{-5}$, $5x^{\frac{2}{3}}$ are not monomials

Polynomial in x:- which have more than one terms like $2x^2 + 3x + 1$ or $2 + \frac{5}{3}x^2$.

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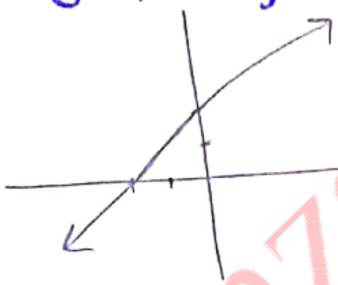
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Lecture 08 Graphs of functions

Defination:- A graph of an equation is just the points on xy-plane that satisfy the equation. Similarly graph of a function f in xy-plane is graph of $y = f(x)$

Example:- Sketch graph of (i) $f(x) = x + 2$

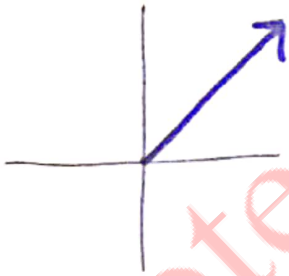
$y = x + 2$



(ii) $g(x) = |x|$

As $|x| = \pm x$

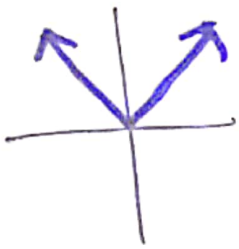
So $g(x) = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$



$g(x) = x$



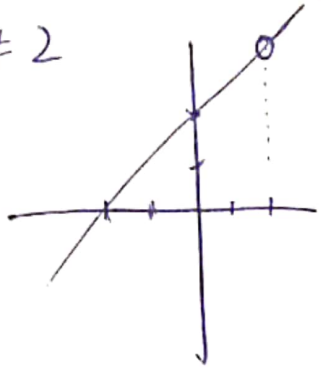
$g(x) = -x$



$g(x) = |x|$

(iii) $f(x) = \dfrac{x^2 - 4}{x - 2} = \dfrac{(x+2)(x-2)}{(x-2)} = (x + 2)$

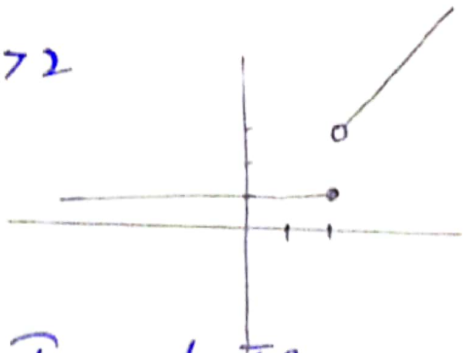
So $y = x + 2, \; x \neq 2$



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(iv) $y = \begin{cases} 1 & \text{if } x \leq 2 \\ x+2 & \text{if } x > 2 \end{cases}$



Eg

Graphing functions by Translations:-

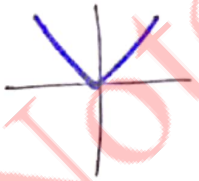
$y = f(x) + c$ graph of $f(x)$ Translates UP by c units.

$y = f(x) - c$ graph of $f(x)$ Translates DOWN by c units.

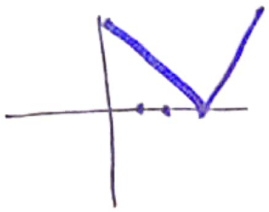
$y = f(x+c)$ graph of $f(x)$ Translates LEFT by c units.

$y = f(x-c)$ graph of $f(x)$ Translates RIGHT by c units.

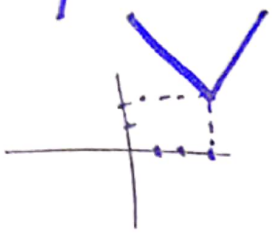
Example:- Graph of $f(x) = |x-3| + 2$



$f(x) = |x|$



$f(x) = |x-3|$



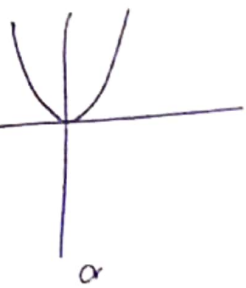
$f(x) = |x-3| + 2$

$y = x^2 - 4x + 5$

$y = x^2 - 4x + 5 + 4 - 4$

$= (x^2 - 4x + 4) + 1$

$= (x-2)^2 + 1$





$f(x) = x^2 - 4x + 5$

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Reflections

- $(-x, y)$ is reflection of (x, y) about y-axis
- $(x, -y)$ is reflection of (x, y) about x-axis.

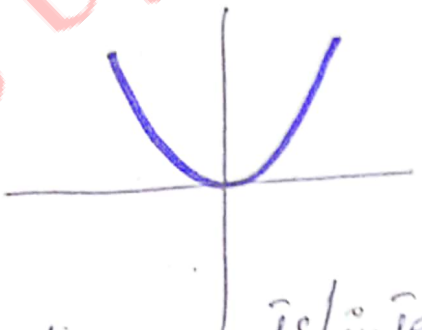
$\Rightarrow$ Graphs of $y = f(x)$ & $y = f(-x)$ are reflections of one another about y-axis.

$\Rightarrow$ Graphs of $y = f(x)$ & $y = -f(x)$ are reflections of one another about x-axis.

Vertical line TEST:-

A graph in the plane is graph of a function if & only if NO VERTICAL line intersects the graph more than once.

For example



Every vertical line cuts/intersects the graph only at one point. So it is graph of a function.

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Lecture 09. Limits

In calculus we find areas of plane regions & find Tangent lines to curves.

⇒ Calculus that comes out of Tangent problem is called differential Calculus.

⇒ Calculus That comes out of Area problems is called Integral Calculus.

Limits:- Limits are basically a way to study the behaviour of y-values of a function in response to x-values as they approach some number or go to infinity.

Imp Formulas:-

Notation	How to read.
$\lim_{x \to x_0^+} f(x) = L_1$	Limit of $f(x)$ as x approaches x_0 from the right is equal L_1.
$\lim_{x \to x_0^-} f(x) = L_2$	Limit of $f(x)$ as x approaches x_0 from the left is equal to L_2.
$\lim_{x \to x_0} f(x) = L$	Limit of $f(x)$ as x approaches x_0 is equal to L.

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Example:- $\lim_{x \to 0} \sin\left(\frac{\pi}{x}\right)$

Its graph has no limiting value as its graph oscillates b/w 1 & -1.



Existence of limit:-

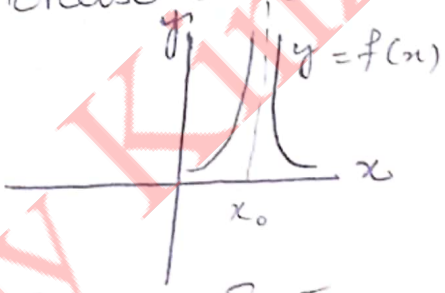
Function does not always have a limit as x value approaches some number. Here we say that limit does not exist. / DNE.

Limit does not exist due to

⟹ Oscillations

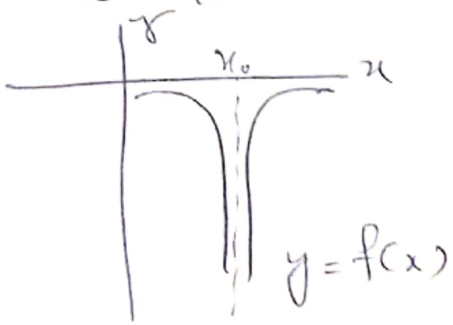
⟹ Unbounded increase or decrease

For example:-



There is increase in y without bound. So limit does not exist due to unboundedness.

In the graph



There is decrease in y without bound

So limit ~~decrease~~ does not exist.

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Lecture 10

Limits & Computational Techniques

Limit:

$$\lim_{x \to a} k = k \quad (\text{e.g } \lim_{x \to 2} 3 = 3)$$

Left hand limit:

$$\lim_{x \to a^-} k = k$$

Right hand limit:

$$\lim_{x \to a^+} k = k$$

Limits of constant functions:-

$$\lim_{x \to a} k = k \quad (\text{e.g } \lim_{x \to 2} 3 = 3)$$

$$\lim_{x \to \infty} k = k \quad (\text{e.g } \lim_{x \to \infty} 3 = 3)$$

$$\lim_{x \to -\infty} k = k \quad (\text{e.g } \lim_{x \to -\infty} 3 = 3)$$

Limits of (variable) functions:-

$$\lim_{x \to a} x = a$$

$$\lim_{x \to +\infty} x = +\infty$$

$$\lim_{x \to -\infty} x = -\infty$$

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Theorem:-

If $L_1 = \lim f(x)$ & $L_2 = \lim g(x)$ both exists

$\Rightarrow \lim[f(x)+g(x)] = \lim f(x) + \lim g(x) = L_1 + L_2$

$\Rightarrow \lim[f(x)-g(x)] = \lim f(x) - \lim g(x) = L_1 - L_2$

$\Rightarrow \lim[f(x)\cdot g(x)] = \lim f(x) \cdot \lim g(x) = L_1 L_2$

$\Rightarrow \lim\left[\frac{f(x)}{g(x)}\right] = \frac{\lim f(x)}{\lim g(x)} = \frac{L_1}{L_2}, \quad (L_2 \neq 0)$

- Limit of the sum is the sum of the limits
- Limit of the difference is difference of limits.
- Similarly for product & division.
- We can also write

$$\lim[f_1(x) + f_2(x) + f_3(x) + \cdots + f_n(x)]$$
$$= \lim f_1(x) + \lim f_2(x) + \lim f_3(x) + \cdots + \lim f_n(x)$$

- If $f_1 = f_2 = \cdots = f_n$ Then $\lim[f(x)]^n = [\lim f(x)]^n$

Results:-

$\Rightarrow \lim_{x\to a}(x^n) = \left[\lim_{x\to a} x\right]^n = a^n$

$\Rightarrow \lim[k\,f(x)] = k \cdot \lim[f(x)]$

A constant can be moved through a limit sign.

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Example:- $\lim_{x \to 5} (x^2 - 4x + 3)$

$$= \lim_{x \to 5} x^2 - \lim_{x \to 5} 4x + \lim_{x \to 5} 3$$

$$= \lim_{x \to 5} x^2 - 4 \lim_{x \to 5} x + \lim_{x \to 5} 3$$

$$= (5)^2 - 4(5) + 3$$

$$= 25 - 20 + 3$$

$$= 5 + 3$$

$$= 8$$

Theorem:- $\lim_{x \to a} P(x) = \lim_{x \to a} (c_0 + c_1 x + \cdots + c_n x^n)$

$$= c_0 + c_1 a + \cdots + c_n a^n$$

$$= P(a)$$

Limits involving $\frac{1}{x}$:-

$$\lim_{x \to 0^+} \frac{1}{x} = \frac{1}{0} = +\infty$$

$$\lim_{x \to 0^-} \frac{1}{x} = \frac{1}{0} = -\infty$$

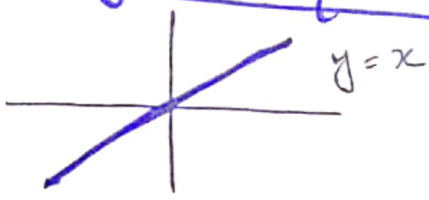
$$\lim_{x \to +\infty} \frac{1}{x} = \frac{1}{+\infty} = 0$$

$$\lim_{x \to -\infty} \frac{1}{x} = \frac{1}{-\infty} = 0$$

Remark:- For every real no. a

$g(x) = \frac{1}{x-a}$ is a Translation of $f(x) = \frac{1}{x}$

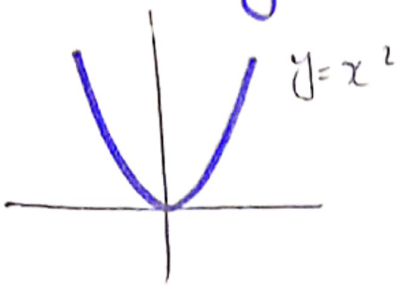
Graph of 1st degree equation:-



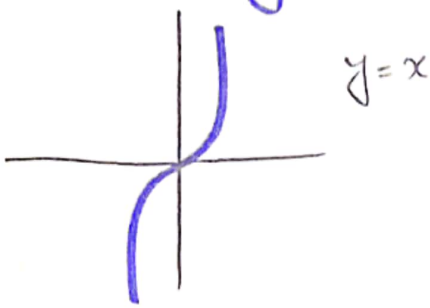
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Graph of 2nd degree equation:-



Graph of 3rd degree equation:-



Example:- $\lim_{x\to+\infty} 2x^5 = 2(+\infty)^5 = +\infty$

$\lim_{x\to+\infty} -7x^6 = -7(+\infty)^6 = -\infty$

$\lim_{x\to+\infty} \frac{1}{x^n} = \left(\lim_{x\to+\infty} \frac{1}{x}\right)^n = \left(\frac{1}{+\infty}\right)^n = (0)^n = 0$

$\lim_{x\to-\infty} \frac{1}{x^n} = \left(\lim_{x\to-\infty} \frac{1}{x}\right)^n = \left(\frac{1}{-\infty}\right)^n = (0)^n = 0$

Rational Function:-

A Rational function is a function defined by the ratio of two polynomials.

Example:-

$$\lim_{x\to2} \frac{5x^3+4}{x-3} = \frac{\lim_{x\to2}(5x^3+4)}{\lim_{x\to2}(x-3)} \Rightarrow \frac{5(2)^3+4}{2-3} \Rightarrow \frac{44}{-1} \Rightarrow -44$$

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Example:- $\lim_{x \to 2} \frac{x^2-4}{x-2}$

$= \lim_{x \to 2} \frac{(x+2)(x-2)}{(x-2)}$

$= \lim_{x \to 2} (x+2) \Rightarrow (2+2) \Rightarrow 4$

Limit of functions of Type $\frac{1}{x}$.

⇒ The limit may be $+\infty$.

⇒ The limit may be $-\infty$.

⇒ The limit may be $+\infty$ from one side & $-\infty$ from other side.

$\lim_{x \to 4^+} \frac{2-x}{(x-4)(x+2)} \Rightarrow \frac{2-4}{(4-4)(4+2)}$

$= \frac{-2}{(0)(6)} \Rightarrow \frac{-2}{0} \Rightarrow -\infty$

Example:- $\lim_{x \to -\infty} \frac{4x^2-x}{2x^3-5} = \lim_{x \to -\infty} \left(\frac{\frac{4x^2}{x^3} - \frac{x}{x^3}}{\frac{2x^3}{x^3} - \frac{5}{x^3}} \right)$

$= \frac{\lim_{x \to -\infty} \frac{4}{x} - \frac{1}{x^2}}{\lim_{x \to -\infty} 2 - \frac{5}{x^3}} \Rightarrow \frac{4 \lim_{x \to -\infty} \frac{1}{x} - \lim_{x \to -\infty} \frac{1}{x^2}}{\lim_{x \to -\infty} 2 - 5 \lim_{x \to -\infty} \frac{1}{x^3}}$

$= \frac{4\left(\frac{1}{\infty}\right) - \frac{1}{\infty^2}}{2 - 5\left(\frac{1}{\infty^3}\right)} \Rightarrow \frac{4(0) - 0}{2 - 5(0)} \Rightarrow \frac{0}{2} = 0$

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Quick Rule:- (only for limits of $+\infty$ & $-\infty$)

$$\lim_{x\to+\infty} \frac{c_0+c_1x+\cdots+c_nx^n}{d_0+d_1x+\cdots+d_nx^n} = \frac{\lim_{x\to\infty} c_nx^n}{d_nx^n}$$

$$\lim_{x\to-\infty} \frac{c_0+c_1x+\cdots+c_nx^n}{d_0+d_1x+\cdots+d_nx^n} = \lim_{x\to-\infty} \frac{c_nx^n}{d_nx^n}$$

Not True if limit $x\to a$.

Example:- $\lim_{x\to+\infty} \frac{4x^2-x}{2x^3-5} = \lim_{x\to+\infty} \frac{4x^2}{2x^3} \Rightarrow \frac{2}{x} \Rightarrow \frac{2}{\infty} \Rightarrow 0$

Lecture 11

Limits: A Rigorous Approach.

$$\lim_{x\to a} f(x) = L$$

The concept of approaches is intuitive. For any number $\varepsilon > 0$ if we can find an open interval (x_0, x_1) on x-axis containing a point "a" such that

$L-\varepsilon < f(x) < L+\varepsilon$ for each x in (x_0, x_1) except possibly $x=a$ Then

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$\lim_{x \to a} f(x) = L$

So $f(x)$ is in interval $(L-\varepsilon, L+\varepsilon)$

Remark:- $\frac{1}{10^{10^{100}}}$ The bottom is called a GOOGOLPLEX

$L-\varepsilon < f(x) < L+\varepsilon$ can also be written as $|f(x)-L| < \varepsilon$ & $(a-\delta, a) \cup (a, a+\delta)$ as $0 < |x-a| < \delta$

Example:- Find ε in $\lim_{x \to 2} (3x-5) = 1$

$|(3x-5)-1| < \varepsilon$ if $0 < |x-2| < \delta$

here $f(x) = 3x-5$, $L = +1$, $a = 2$

we have to find ε.

$|3x-5-1| < \varepsilon$

$|3x-6| < \varepsilon$

$3|x-2| < \varepsilon$

$|x-2| < \frac{\varepsilon}{2}$ if $0 < (x-2) < \frac{\varepsilon}{3}$

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Lecture 12 Continuity.

Continuous Function:-

Function $f(x)$ is continuous if (i) $f(c)$ is defined. (ii) $\lim_{x\to c} f(x)$ exists. (iii) $\lim_{x\to c} f(x) = f(c)$

If any of these conditions fails to hold for a function then $f(x)$ is called discontinuous.

c is called point of discontinuity.

If function is continuous on all points in (a,b) then it is called continuous over (a,b)

If function is continuous on $(-\infty, +\infty)$ then its called continuous function.

Example:- $f(x) = \dfrac{x^2-4}{x-2}$ it is discontinuous at $x=2$ because $f(2)$ is undefined.

Example:- $g(x) = \begin{cases} \dfrac{x^2-4}{x-2} & \text{if } x \neq 2 \\ 3 & \text{if } x = 2 \end{cases}$

$g(x)$ is discontinuous because

$$\lim_{x\to 2} g(x) = \lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} \frac{(x+2)(x-2)}{(x-2)}$$

$$= \lim_{x\to 2} (x+2) = 2+2 = 4$$

but in $g(x)$, $g(2) = 3$. That's not True.

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Show That $f(x) = x^2 - 2x + 1$ is continuous.

$$\lim_{x \to c} x^2 - 2x + 1 = c^2 - 2c + 1$$

function is continuous.

Theorem:-

Polynomials are continuous functions.

Example:-

$f(x) = |x|$ is continuous or not?

$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$\lim_{x \to c} f(x) = f(c)$

Let $c \geq 0$ Then $f(c) = c$

$$\lim_{x \to c} f(x) = \lim_{x \to c} |x| = \lim_{x \to c} x = c$$

Let $c \leq 0$ Then $f(c) = -c$

$$\lim_{x \to c} f(x) = \lim_{x \to c} |x| = \lim_{x \to c} (-x) = -(-c) = c$$

So function is continuous.

Properties:-

If f & g are continuous, Then

a) $f+g$ is continuous at c.

b) $f-g$ is continuous at c.

c) $f.g$ is continuous at c.

d) f/g is continuous at c if $g(c) \neq 0$ & dis continuous if $g(c) = 0$

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Continuity of Rational functions:-

Example:- $h(x) = \dfrac{x^2-9}{x^2-5x+6}$

we'll find factors of denominator.

$$x^2-5x+6 = x^2-(3+2)x+6$$
$$= x^2-3x-2x+6$$
$$= x(x-3)-2(x-3)$$
$$= (x-2)(x-3)$$

So $h(x) = \dfrac{x^2-9}{(x-2)(x-3)}$

So $h(x)$ is discontinuous at $x=2$ & $x=3$.

Theorem:-

If $\lim g(x) = L$ & if the function f is continuous at L, Then $\lim f(g(x)) = f(L)$

(i-e) $\lim f(g(x)) = f(\lim g(x))$.

Example:- $f(x) = \lim_{x\to 3} |5-x^2|$

Here $f(x) = |x|$, $g(x) = 5-x^2$

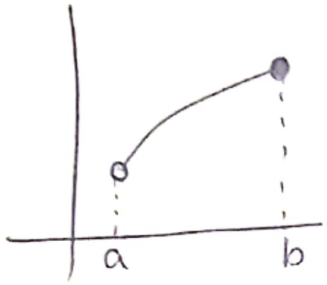
So, as

$$\lim_{x\to 3} f(g(x)) = \cancel{\lim_{x\to 3} f}$$
$$= \lim_{x\to 3} |5-x^2| \Rightarrow \left|\lim_{x\to 3}(5-x^2)\right|$$
$$= |5-3^2| \Rightarrow |5-9| = |-4| = 4$$

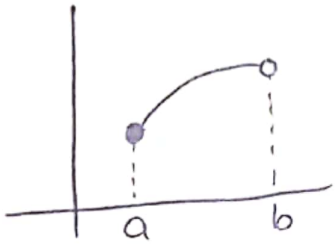
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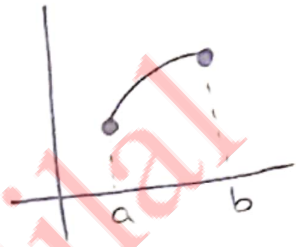
Theorem:- If function g is continuous at point c & function f is continuous at point $g(c)$ Then composition $f(g(x)) = fog(x)$ is continuous.



discontinuous at "a"



discontinuous at "b".



continuous at a & b.

Theorem:-

1. A function is called continuous from the left at point c if
   a) $f(c)$ is defined.
   b) $\lim_{x \to c^-} f(x)$ exists.
   c) $\lim_{x \to c^-} f(x) = f(c)$

2. A function is called continuous from the right at point c if
   a) $f(c)$ is defined.
   b) $\lim_{x \to c^+} f(x)$ exists.
   c) $\lim_{x \to c^+} f(x) = f(c)$

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Theorem:- $f(x)$ is continuous on $[a,b]$ (closed interval) if

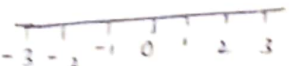
1. f is continuous on (a,b)
2. f is continuous from left on (a,b)
3. f is continuous from right on (a,b).

Example:- Show that $f(x) = \sqrt{9-x^2}$ is continuous on $[3,-3]$.

1. $\lim_{x\to c} f(x) = \lim_{x\to c} \sqrt{9-x^2} = \sqrt{\lim_{x\to c}(9-x^2)} = \sqrt{9-c^2} = f(c)$
2. $\lim_{x\to 3^-} f(x) = \lim_{x\to 3^-} \sqrt{9-x^2} = \sqrt{\lim_{x\to 3^-}(9-x^2)} = \sqrt{9-3^2} = 0 = f(3)$
3. $\lim_{x\to -3^+} f(x) = \lim_{x\to -3^+} \sqrt{9-x^2} = \sqrt{\lim_{x\to -3^+}(9-x^2)} = \sqrt{9-(-3)^2} = 0 = f(-3)$

(We approach 3 from left & -3 from right)

So f is continuous on $[-3,3]$.



Intermediate value Theorem:-

If f is continuous on closed interval $[a,b]$ & c is any number b/w $f(a)$ & $f(b)$ inclusive, Then There is at least one number x in interval $[a,b]$ such that $f(x)=C$.

If f is continuous on $[a,b]$ & if $f(a)$ & $f(b)$ have opposite signs Then There is at least one solution of $f(x)=0$ in interval (a,b).

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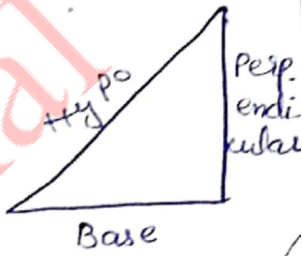
Lecture 13

limits & continuity of Trigonometric functions

Continuity of sine & cosine:-

As $\sin\theta = \dfrac{\text{opposite side / Perpendicular / altitude}}{\text{Hypotaneous}}$

$\cos\theta = \dfrac{\text{Adjacent side / Base}}{\text{Hypotaneous}}$



Theorem:-

The functions $\sin x$ & $\cos x$ are continuous.

$$\lim_{x\to 0} \sin x = 0 \quad \& \quad \lim_{x\to 0} \cos x = 1$$

- let $h = x - c$, so $x = h + c$ Then $x \to c$ is equivalent to $h \to 0$, so

function is continuous at c if

1. $f(c)$ is defined.
2. $\lim_{h\to 0} f(h+c)$ exists.
3. $\lim_{h\to 0} f(h+c) = f(c)$

Proof:- $\lim_{h\to 0} \sin(c+h) = \sin c$

$$\lim_{h\to 0} \sin(c+h) = \lim_{h\to 0} \left[\sin(c)\cos(h) + \cos(c)\sin(h)\right]$$

$$= \lim_{h\to 0} \sin(c)\cos(h) + \lim_{h\to 0} \cos(c)\sin(h)$$

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$$= \sin c \lim_{h\to 0} \cos(c) + \cos(c) \lim_{h\to 0} \sin(c)$$

$$= \sin(c)(1) + \cos(c)(0)$$

$$= \sin(c) + 0$$

$$\lim_{h\to 0} \sin(c+h) = \sin(c)$$

Similarly we can do proof for $\lim_{h\to 0} \cos(c+h) = \cos(c)$

Continuity for other Trigonometric functions:-

If $f(x)$ & $g(x)$ are continuous then $h(x) = \frac{f(x)}{g(x)}$ is also continuous. (except $g(x) = 0$)

Similarly $\sin x$ & $\cos x$ are continuous, so

$\tan(x) = \frac{\sin x}{\cos x}$ is also continuous. (except $\cos x = 0$)

Like wise

$\cot(x) = \frac{\cos(x)}{\sin(x)}$, $\sec(x) = \frac{1}{\cos(x)}$, $\operatorname{cosec}(x) = \frac{1}{\sin(x)}$

Squeeze Theorem:-

$$\lim_{x\to 0} \frac{\sin(x)}{x} = 1, \quad \lim_{x\to 0} \frac{1-\cos(x)}{x} = 0$$

Squeezing Theorem:-

Let f, g & h be functions satisfying $g(x) \le f(x) \le h(x)$ for all x in some open interval containing point a, with possible exception that inequality need not to hold at a.

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$$\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L$$

Then f also has this limit as x approaches to a $\lim_{x\to a} f(x) = L$

Example:-

$$\lim_{x\to 0} x^2 \sin^2\left(\frac{1}{x}\right)$$

Remember $0 \le \sin(x) \le 1$

So $0 \le \sin^2(x) \le 1$

Also $0 \le \sin^2\left(\frac{1}{x}\right) \le 1$

multiply by x^2 on all sides

$$(0)(x^2) \le x^2 \sin^2\left(\frac{1}{x}\right) \le (1)(x^2)$$

$$0 \le x^2 \sin^2\left(\frac{1}{x}\right) \le x^2$$

But $\lim_{x\to 0} 0 = \lim_{x\to 0} x^2 = 0$

Both L.H.S & R.H.S are zero So by Squeezing Theorem

$$\lim_{x\to 0} x^2 \sin^2\left(\frac{1}{x}\right) = 0$$

Theorem:-

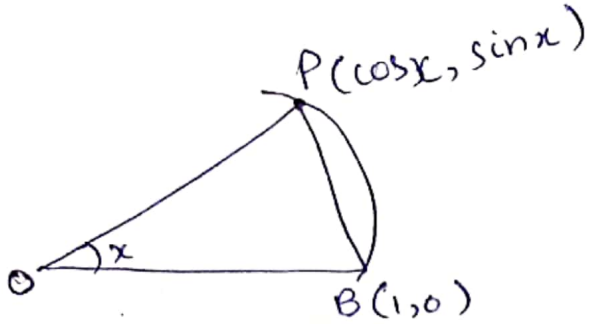
$$\lim_{x\to 0} \frac{\sin(x)}{x} = 1$$

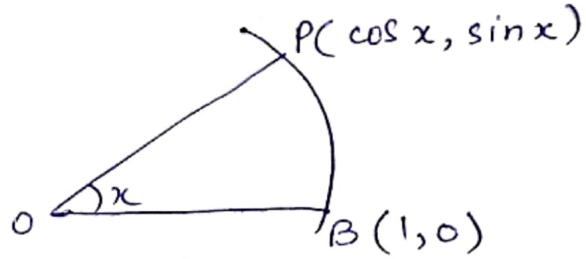
Proof:-

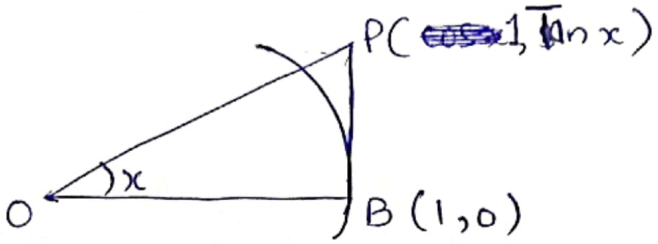
let $0 < x < \frac{\pi}{2}$ & take the following shapes.

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$0 <$ area of $\Delta OBP <$ area of sector $OBP <$ area of ΔOBP

$$0 < \frac{1}{2}(\text{base})(\text{height}) < \frac{1}{2}r^2\theta < \frac{1}{2}(\text{base})(\text{height})$$

$$0 < \frac{1}{2}(1)(\sin x) < \frac{1}{2}(1)^2(x) < \frac{1}{2}(1)(\tan x)$$

$$0 < \frac{1}{2}\sin x < \frac{1}{2}x < \frac{1}{2}\tan x$$

multiplying by $\frac{2}{\sin x}$

$$0 < \frac{1}{2}\sin x\left(\frac{2}{\sin x}\right) < \frac{1}{2}x\left(\frac{2}{\sin x}\right) < \frac{1}{2}\tan x\left(\frac{2}{\sin x}\right)$$

$$0 < 1 < \frac{x}{\sin x} < \frac{\sin x}{\cos x}\cdot\frac{1}{\sin x}$$

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x}$$

Take Reciprocal

$$1 > \frac{\sin x}{x} > \cos x$$

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As $0 < x < \frac{\pi}{2}$

$1 < \frac{\sin x}{x} < \cos x$ becomes

As $\lim_{x \to 0} \cos x = 1$ & $\lim_{x \to 0} 1 = 1$

Take limits

$$\lim_{x \to 0} 1 < \lim_{x \to 0} \frac{\sin x}{x} < \lim_{x \to 0} \cos x$$

$$1 < \lim_{x \to 0} \frac{\sin x}{x} < 1$$

By Squeezing Theorem

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

Lecture 14

Tangent lines & rates of Change

Limit of a secant line (Slope)

$$m_{sec} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Slope of secant line will approach that of Tangent line at P.

$$m_{Tan} = \lim_{x_1 \to x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Also

$$\text{Average velocity} = \frac{\text{distance}}{\text{Time}} = \frac{d_1 - d_0}{t_1 - t_0} = \frac{f(t_1) - f(t_0)}{t_1 - t_0}$$

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Defination:-

If $y=f(x)$, then average rate of change of y w.r.t x over the interval $[x_0, x_1]$ is slope m_{sec} of secant line joining the points $(x_0, f(x_0))$ & $(x_1, f(x_1))$ on graph of f.

$$m_{sec} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Defination:- If $y=f(x)$ then instataneous rate of change of y with respect to x at point x_0 is the slope

$$m_{Tan} = \lim_{x_1 \to x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Example:- $y = f(x) = x^2+1$

a) Find average rate of y w.r.t x over $[3,5]$

b) Find instataneous rate of y w.r.t x at $x=x_0$ point $x_0 = -4$

c) Find instataneous rate of change of y w.r.t x at a general point.

a) $y = f(x) = x^2+1$, $x_0 = 3$, $x_1 = 5$

$$m_{sec} = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{f(5) - f(3)}{5-3} = \frac{(5^2+1) - (3^2+1)}{2}$$

$$= \frac{26-10}{2} = \frac{16}{2} = 8$$

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(b) $y = f(x) = x^2 + 1 \quad , \quad x_0 = -4$

$$m_{Tan} = \lim_{x_1 \to x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \lim_{x_1 \to x_0} \frac{x_1^2 + 1 - (x_0^2 + 1)}{x_1 - x_0}$$

$$= \lim_{x_1 \to x_0} \frac{x_1^2 + 1 - (-4)^2 - 1}{x_1 - (-4)} = \lim_{x \to -4} \frac{x_1^2 + 1 - 16 - 1}{x_1 + 4}$$

$$= \lim_{x_1 \to -4} \frac{x_1^2 - 16}{x_1 + 4} = \lim_{x \to -4} \frac{(x_1 + 4)(x_1 - 4)}{(x_1 + 4)}$$

$$= \lim_{x_1 \to -4} (x_1 - 4) = -4 - 4 = -8$$

(c)

$$m_{Tan} = \lim_{x_1 \to x_0} \frac{f(x_1) f(x_0)}{x_1 - x_0}$$

$$= \lim_{x_1 \to x_0} \frac{(x_1^2 + 1) - (x_0^2 - 1)}{x_1 - x_0} = \lim_{x_1 \to x_0} \frac{x_1^2 + 1 - x_0^2 - 1}{x_1 - x_0}$$

$$= \lim_{x_1 \to x_0} \frac{x_1^2 - x_0^2}{x_1 - x_0} = \lim_{x_1 \to x_0} \frac{(x_1 + x_0)(x_1 - x_0)}{x_1 - x_0}$$

$$= \lim_{x_1 \to x_0} (x_1 + x_0) = (x_0 + x_0) = 2x_0$$

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Lecture 15

The Derivative

We can rewrite $m_{Tan} = \lim_{x_1 \to x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0}$

as

$h = x_1 - x_0$, $x_1 = x_0 + h$ & $h \to 0$ as $x_1 \to x_0$

$$m_{Tan} = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

If $P(x_0, y_0)$ is a point on graph of a function f Then Tangent line to graph of f at P is defined to be line Through P. with Slope.

$$m_{Tan} = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

Example:- Find slope & equation of Tangent line to graph of $f(x) = x^2$ at point $(3, 9)$

$f(x) = x^2$, $x_0 = 3$, $y_0 = 9$

$$m_{Tan} = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

$$= \lim_{h \to 0} \frac{f(3 + h) - f(3)}{h}$$

$$= \lim_{h \to 0} \frac{(3+h)^2 - 9}{h}$$

$$= \lim_{h \to 0} \frac{9 + h^2 + 6h - 9}{h}$$

$f(x) = x^2$

$f(x_0 + h) = (x_0 + h)^2$

$= (3 + h)^2$

& $f(x_0) = x_0^2$

$= (3)^2$

$= 9$

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$$= \lim_{h \to 0} \frac{h^2 + 6h}{h}$$

$$= \lim_{h \to 0} \frac{h(h+6)}{h}$$

$$= \lim_{h \to 0} (h+6) = 0+6 = 6$$

Equation of Tangent line

$$y - y_0 = m(x - x_0)$$

$$y - 9 = 6(x-3)$$

$$y - 9 = 6x - 18$$

$$y = 6x + 9 - 18$$

$$y = 6x - 9$$

Defination:- The function f is denoted by formula $f' = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ is called with respect to x of function f. The domain of f consists of all x for which limit exists.

⇒ We can interpret derivative in 2-ways.

- Geometric interpretation of derivative
- Rate of change is an interpretation of derivative also.

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Example:- $f(x) = x^2 + 1$

Find $f'(x)$.

$$f' = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$f(x+h) = (x+h)^2 + 1$$
$$= x^2 + h^2 + 2hx + 1$$

put values

$$f'(x) = \lim_{h \to 0} \frac{x^2 + h^2 + 2hx + 1 - (x^2 + 1)}{h}$$

$$= \lim_{h \to 0} \frac{x^2 + h^2 + 2hx + 1 - x^2 - 1}{h}$$

$$= \lim_{h \to 0} \frac{h^2 + 2hx}{h}$$

$$= \lim_{h \to 0} \frac{h(h + 2x)}{h}$$

$$= \lim_{h \to 0} (h + 2x)$$

$$= 0 + 2x$$

$$f'(x) = 2x$$

$$f'(2) = 2(2) = 4$$

$$f'(-2) = 2(-2) = -4$$

$$f'(0) = 2(0) = 0$$

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Example:- $f(x) = \sqrt{x}$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$f(x+h) = \sqrt{x+h}$$

$$f'(x) = \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

$$= \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \times \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$$

$$= \lim_{h \to 0} \frac{(\sqrt{x+h})^2 - (\sqrt{x})^2}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \lim_{h \to 0} \frac{x + h - x}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \lim_{h \to 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})}$$

$$= \lim_{h \to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}}$$

$$= \frac{1}{\sqrt{x+0} + \sqrt{x}} = \frac{1}{\sqrt{x} + \sqrt{x}} = \frac{1}{2\sqrt{x}}$$

Derivative Notation:-

The process of finding the derivative is called Differentiation.

It is written as $\frac{d}{dx}[f(x)]$ read as "The derivative of f w.r.t x.

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$\frac{d}{dx}[f(x)]$ can also be written as $f'(x)$.

$$Y = f(x)$$

$$\frac{d}{dx}(y) = \frac{d}{dx}[f(x)]$$

$$\frac{dy}{dx} = f'(x)$$

Example:- $f(x) = \sqrt{x}$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

or $$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}$$

Existence of derivative:-

⇒ Derivative exists at points where limit exists.

⇒ f is differentiable on an open interval (a, b) if it is differentiable at each point in (a, b).

⇒ f is not differentiable, we say derivative does not exists.

Non-differentiability occurs when graph of $f(x)$ has

a) corners
b) vertical Tangents
c) Points of discontinuity.

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Theorem:- If f is differentiable at a point x_0 Then f is also continuous at x_0.

Proof:-

$$\lim_{h\to 0} f(x_0+h) = f(x_0)$$

$$\lim_{h\to 0} \left(f(x_0+h) - f(x_0)\right) = 0$$

L.H.S

$$\lim_{h\to 0} \left(f(x_0+h) - f(x_0)\right)$$

= multiply & divide by "h"

$$= \lim_{h\to 0} \left(\left[\frac{f(x_0+h) - f(x_0)}{h} \right] \cdot h \right)$$

$$= \lim_{h\to 0} \left[\frac{f(x_0+h) - f(x_0)}{h} \right] \cdot \lim_{h\to 0} h.$$

$$= f'(x_0) \cdot 0$$

$$= 0 = \text{R.H.S.}$$

Example:- $f(x) = |x|$, $f'(x) = ?$

$$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$f'(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

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